

E_macro — Master Findings Report

Consolidated Synthesis Across Sources A–F

Macro Embeddings project — internal technical report

Prepared 2026-07-16 · Data as of 2026-07-14

1. Executive Summary

E_macro is a six-pillar county-level feature set (Sources A–F) built to characterize the economic identity of all 3,144 US counties/county-equivalents from six independent public data sources: Wikipedia intro-text embeddings (A), BLS industry location quotients (B), FRED unemployment/GDP velocity (C), BTS freight flows (D), IRS capital composition (E), and USDA structural typology (F). Every source reached full or near-full county coverage (typically 99.97%–100%), and every source's findings report includes an explicit proposal-alignment assessment against E_macro_extendedProposal.pdf and macro_pre_scoping_spec.pdf.

Three findings matter most across the whole project:

- 1 Structural pillars explain each other far better than any of them explains short-run economic momentum.** Every test run against Source C's 3-year velocity measure — across Sources A, D, and F — tops out at $|r| \leq 0.08$. The two direct pillar-to-pillar tests run this round (Source D × Source F, Source A × Source F) produce the two strongest, most significant effects in the entire project: $r = -0.22$ (demographic distress vs. freight tonnage) and $r = 0.25$ (metro status vs. Wikipedia text length) — roughly 3-6x larger than any momentum link. E_macro's pillars read as a mosaic of complementary structural signals, not a set of momentum predictors.
- 2 Both synergies the proposal specifically describes hold up, at the level the evidence supports.** Trade Logistics (Source D explained by Source F's typology) and Capital Flow (Source F anchoring counties where Source A's text is sparse) are both confirmed at moderate-to-strong strength — but the finer-grained version of Capital Flow (graded distress, not just metro/nonmetro) does not hold, and neither synergy was anticipated to be **stronger** than the momentum links the proposal emphasizes more heavily.
- 3 A few proposal assumptions don't survive contact with the real data,** each independently caught and documented rather than patched over: Source A's intro text carries no economic-transition narrative (its strongest structure is a Wikipedia editorial-convention artifact); Source C's GDP velocity metric measures economy size more than acceleration rate; Source D's product ships tonnage only, no dollar value; Source E has no suppression flag at all (unlike Source B's, which does); and two unrelated federal agencies (BEA/FRED for Source C, USDA/ERS for Source F) independently show the same data-lag gap for Connecticut's 2022 Planning-Region transition.

The following sections summarize each source in turn, then the cross-source synthesis, a consolidated data-quality and proposal-alignment picture, and prioritized next actions.

2. Project Scope & Methodology

`E_macro` organizes its six sources into pillars: Source A (narrative identity), Source B (industrial core), Source C (economic velocity), Source D (trade/movement), Source E (capital flow), and Source F (structural resilience baseline). All six join on `fips_code` against a shared `county_crosswalk.parquet` covering all 3,144 US counties/county-equivalents. Every findings report follows the same structure — executive summary, data & setup, main findings, figures, limitations, next actions, and a proposal-alignment assessment — and every cross-source correlation test in this project uses the same statistical protocol: Pearson r (or a Mantel test for matrix-vs-matrix comparisons) with a 499-permutation significance test, seed 42, so effect sizes are comparable across pillar pairs even though sample sizes differ slightly by pillar (Source A applies its own stub-county and non-50-state filters; the other five sources are at or near full 3,144-county coverage).

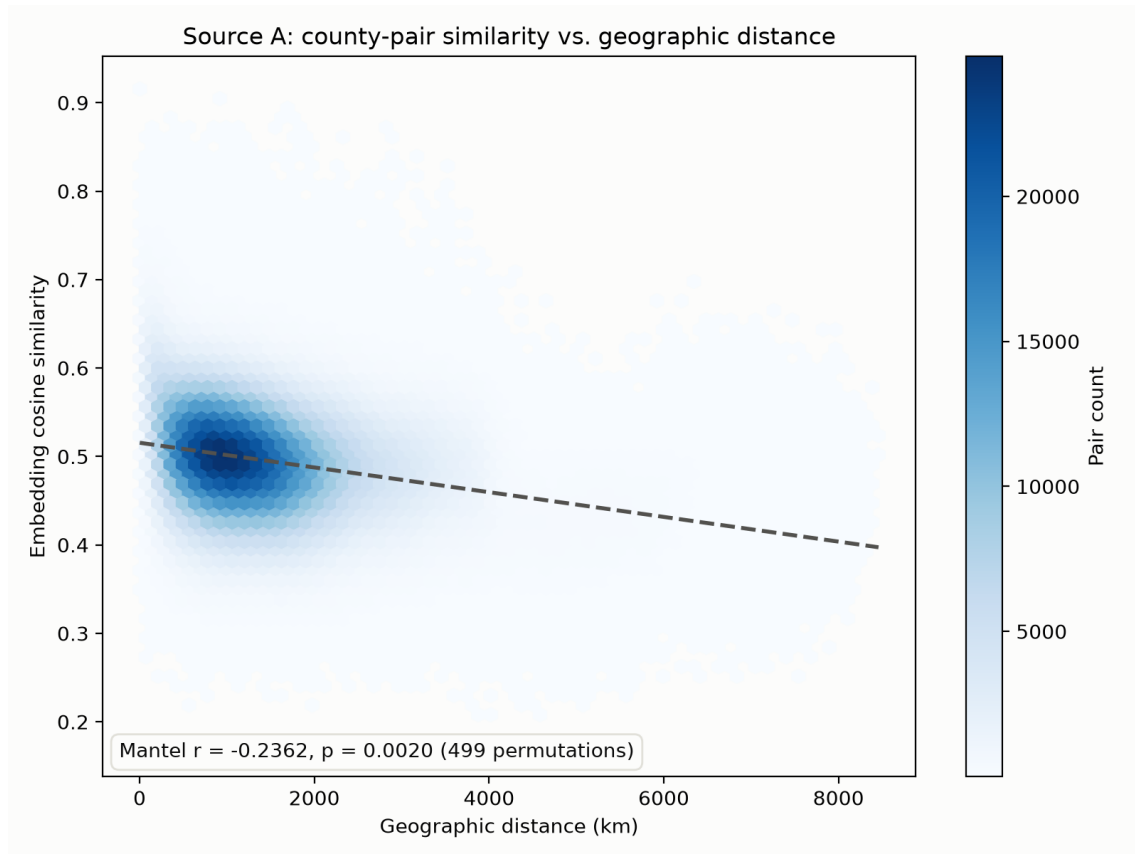
A size-normalization convention recurs throughout: Source C's raw `gdp_velocity` is dollar-denominated and highly correlated with economy size (the five largest values belong to the five largest county economies), so every cross-source test against GDP velocity uses `gdp_velocity_pct = gdp_velocity / gdp_latest` instead of the raw column — a fix Source C's own report recommended and every downstream cross-validation adopted.

3. Per-Source Findings

3.1 Source A — Wikipedia Intro-Text County Embeddings (Narrative Identity)

Source A embeds each county's Wikipedia intro text (`BAAI/bge-m3`, LLM-cleaned to strip boilerplate) and looks for geographic and economic structure in the resulting 1024-dim vectors. Coverage is complete (3,144/3,144). A Mantel test finds a weak but real negative association between geographic distance and embedding similarity ($r = -0.24$, $p = 0.002$), roughly doubled by adopting LLM-cleaned text over a regex baseline. PCA's first component explains only 3.9% of variance and, contrary to an earlier reading, does carry a real theme on the LLM-cleaned embeddings — but that theme is a Wikipedia editorial-convention artifact (Texas-concentrated founding/namesake narrative vs. bare demographic-stub content), not economic-transition language. K-means finds a highly reproducible $k=2$ split ($ARI \approx 0.98$ across seeds) whose geographic coherence is statistically real (permutation $p \leq 0.001$) but small (silhouette ≤ 0.029) — not evidence of clean regional structure. Expanding the embedded text beyond the intro section (more sections, an Economy-only section, with or without LLM cleaning) made counties more similar, not less, and was closed as a dead end.

Proposal alignment: the proposal's central claim for Source A — that intro text carries distinctive economic-transition narrative giving "immediate semantic separation" between counties — is not supported. The strongest structure found is a non-economic editorial artifact, and (per §4 below) embedding distance also shows only a negligible association with real economic variables once tested directly.

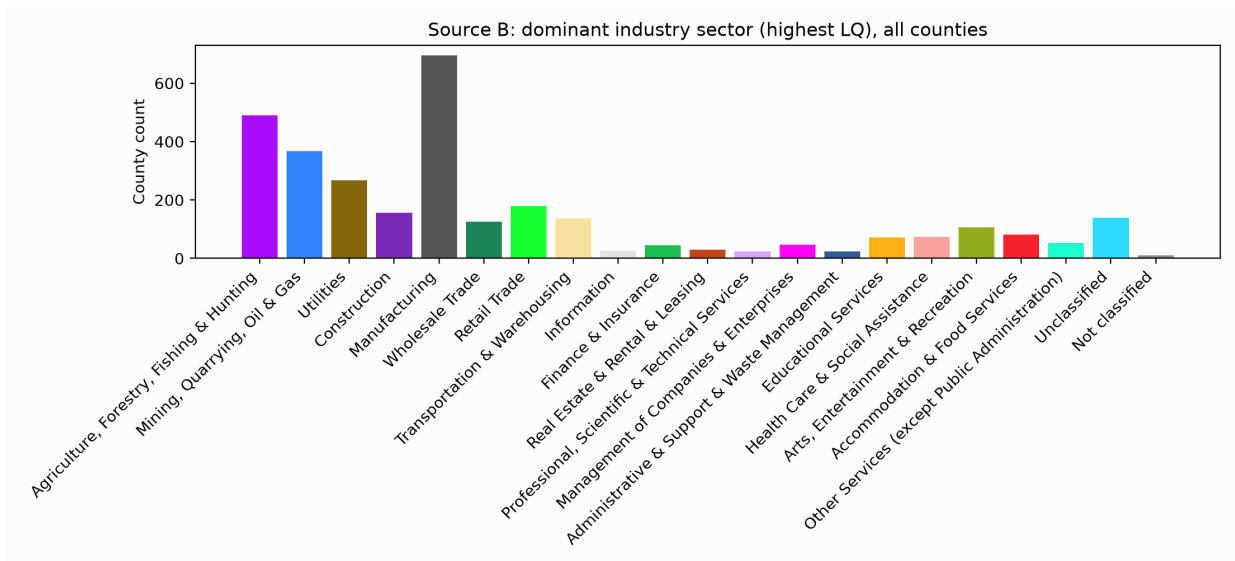


Embedding similarity vs. geographic distance across all county pairs. The shallow trend line visualizes just how weak the geographic signal is (Mantel $r = -0.24$) — this is Source A's single strongest piece of structure.

3.2 Source B — BLS QCEW Location Quotients (Industrial Core)

Source B pulls BLS's pre-computed Location Quotients across the 20 primary NAICS-2 sectors for Private ownership, 2025 Q4. Coverage is 3,143/3,144 (only Kalawao County, HI is absent from BLS's own county-level file). Collapsing each county to its single highest-LQ sector shows production/extraction sectors dominate: Manufacturing (22.1%), Agriculture (15.6%), Mining (11.7%), and Utilities (8.5%) together account for 57.9% of all counties' dominant sector — the opposite emphasis from Source F's typology, where "Nonspecialized" is the largest single category. The most extreme specialization values (Eureka County, NV at LQ 248 in Mining; Surry County, VA at LQ 102 in Utilities, home to a nuclear plant) check out against real-world geography, confirming the metric works as intended. BLS suppresses 30.0% of county×sector cells for small-employer privacy; a suppression-handling bug (BLS ships a suppressed cell as a literal 0, not blank) was caught and fixed during verification, and Source B ships honest nulls rather than attempting a matrix-completion fix that testing showed doesn't meaningfully beat a naive baseline. Cross-validated against Source C, a county's raw specialization **magnitude** is only weakly related to GDP/unemployment velocity, but **which** sector dominates matters more — Real Estate ($r=0.206$), Construction ($r=0.154$), and Wholesale Trade ($r=-0.157$) all survive FDR correction.

Proposal alignment: supported — the mechanism works exactly as described, and the per-sector correlation result is direct evidence for the proposal's claim that Source B distinguishes "what kind" of growth a county experiences, not just how much.

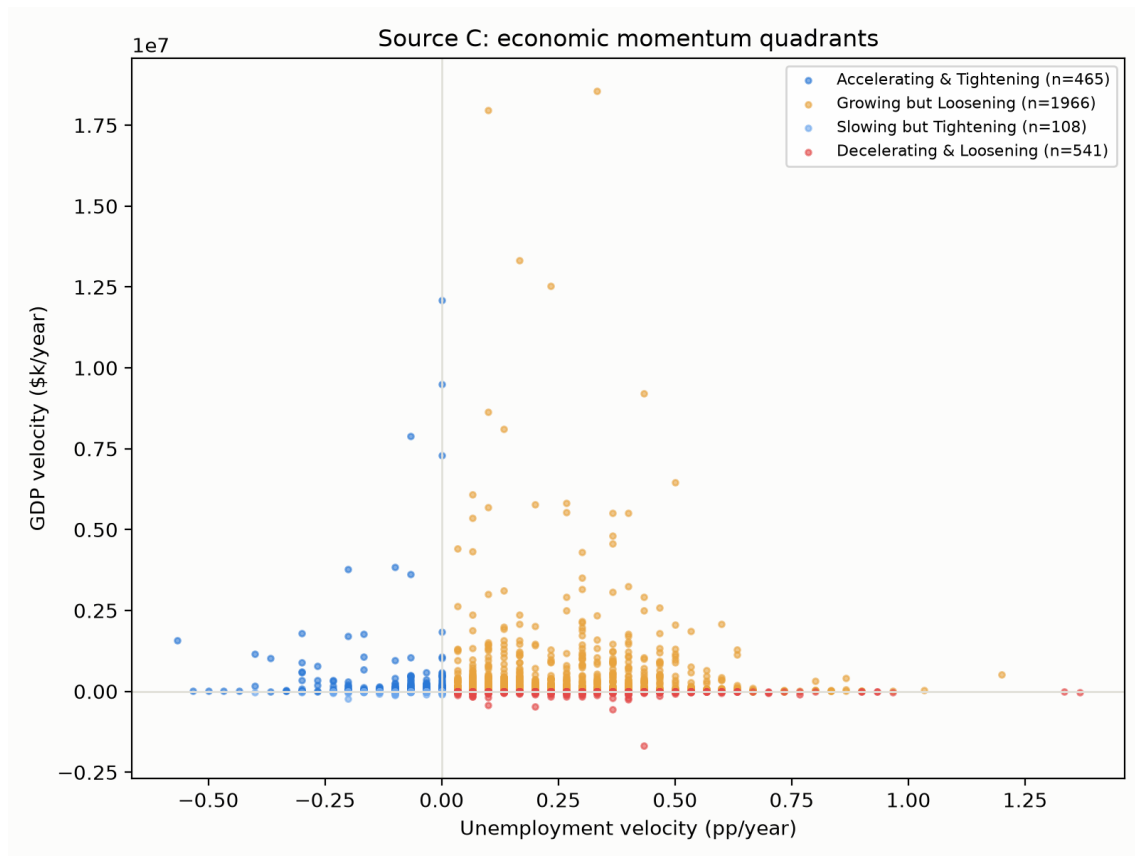


Dominant NAICS-2 sector breakdown across all 3,144 counties. Manufacturing, Agriculture, Mining, and Utilities together account for 57.9% of counties' dominant sector — production/extraction, not diversified services.

3.3 Source C — FRED Time-Series Slope Derivatives (Economic Velocity)

Source C computes each county's rolling 3-year first derivative of unemployment rate and real GDP from FRED's annual series. Coverage is 98.0% full, 2.0% partial (unemployment only — Virginia independent cities and Connecticut Planning Regions account for nearly all of the GDP gap), and one full failure (Kalawao County, HI). The dominant pattern is a broad, shared-direction shift rather than county-specific divergence: 63.8% of counties fall in the "Growing but Loosening" quadrant (GDP still expanding, unemployment still rising off a low base), and the two velocity axes are essentially uncorrelated cross-sectionally ($r = 0.009$) — a level shift shared across nearly the whole country, not the differentiated "different markets, different quadrants" picture the proposal's narrative example implies. A real metric-design limitation: GDP velocity is measured in absolute dollars, so it is dominated by the largest metro economies (King, Harris, Maricopa, LA, New York counties) rather than by counties with unusually sharp *percentage* acceleration — flagged rather than silently fixed, with a percentage-based variant (`gdp_velocity_pct`) recommended and subsequently adopted by every downstream cross-source test.

Proposal alignment: partially supported — the quadrant mechanism does differentiate counties, but a shared macro trend (not idiosyncratic local momentum) explains most of the variation; the proposal's synergy claim with Source B is now testable and is addressed in §4.

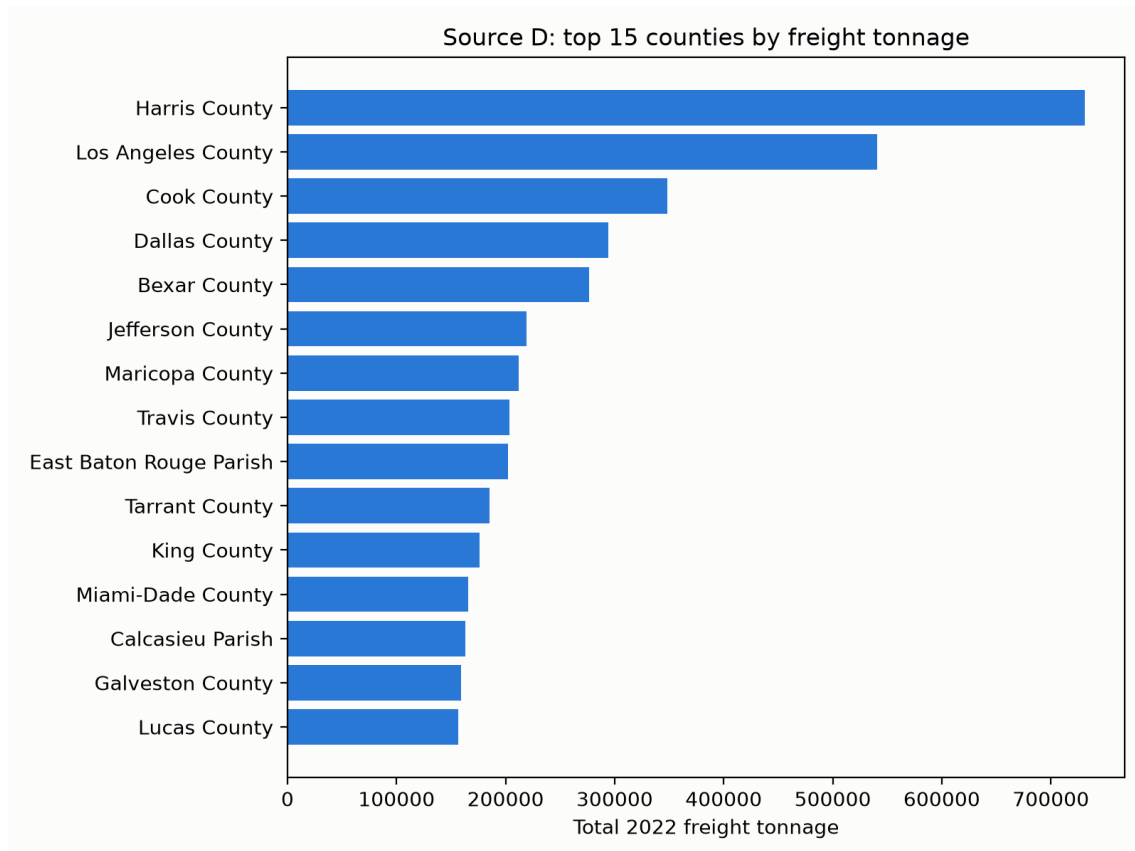


Momentum-quadrant scatter, all 3,080 fully-covered counties, colored by quadrant. One quadrant ("Growing but Loosening") clearly dominates by point density — visual evidence for the shared-macro-trend finding.

3.4 Source D — BTS FAF5 County Commodity Flows (Trade Gravity / Movement)

Source D ingests 2022 county-level freight tonnage from BTS's experimental FAF5 product. Coverage is complete (3,144/3,144, zero nulls). Scalar tonnage alone separates hubs from sinks across roughly two orders of magnitude — Harris County, TX (Houston) leads at 731K tons, while Loving County, TX (population ~64, a genuine Permian Basin oil/gas export point) shows a correctly-captured export imbalance despite being nearly uninhabited. A design-spike finding (hub counties look *less* concentrated than rural ones at a two-state regional scale) reverses at full national scale: nationally, big hubs funnel volume through fewer dominant interstate corridors ($r = 0.278$ between log-tonnage and partner HHI) — a genuine scale-dependent reversal, not a contradiction between rounds of testing. Against Source C, trade volume tracks GDP momentum weakly but monotonically (mean GDP velocity rises from 1.6% to 2.4%/year across tonnage quartiles) — modest, but more consistent than Source F's near-zero momentum link. Two real scope deviations from the proposal: BTS's own county/FAF-zone hybrid table structure already solves the "combinatorial matrix explosion" both scoping docs worried about (superseding a proposed top-K truncation fix), and the experimental county-level product ships tonnage only, no dollar value, despite both docs describing Source D as tracking both.

Proposal alignment: supported at national scale — the core framing (distinguishing pass-through corridors and exporters from consumer sinks) holds, with two documented scope deviations and one new finding (the HHI reversal) neither scoping doc could have anticipated.

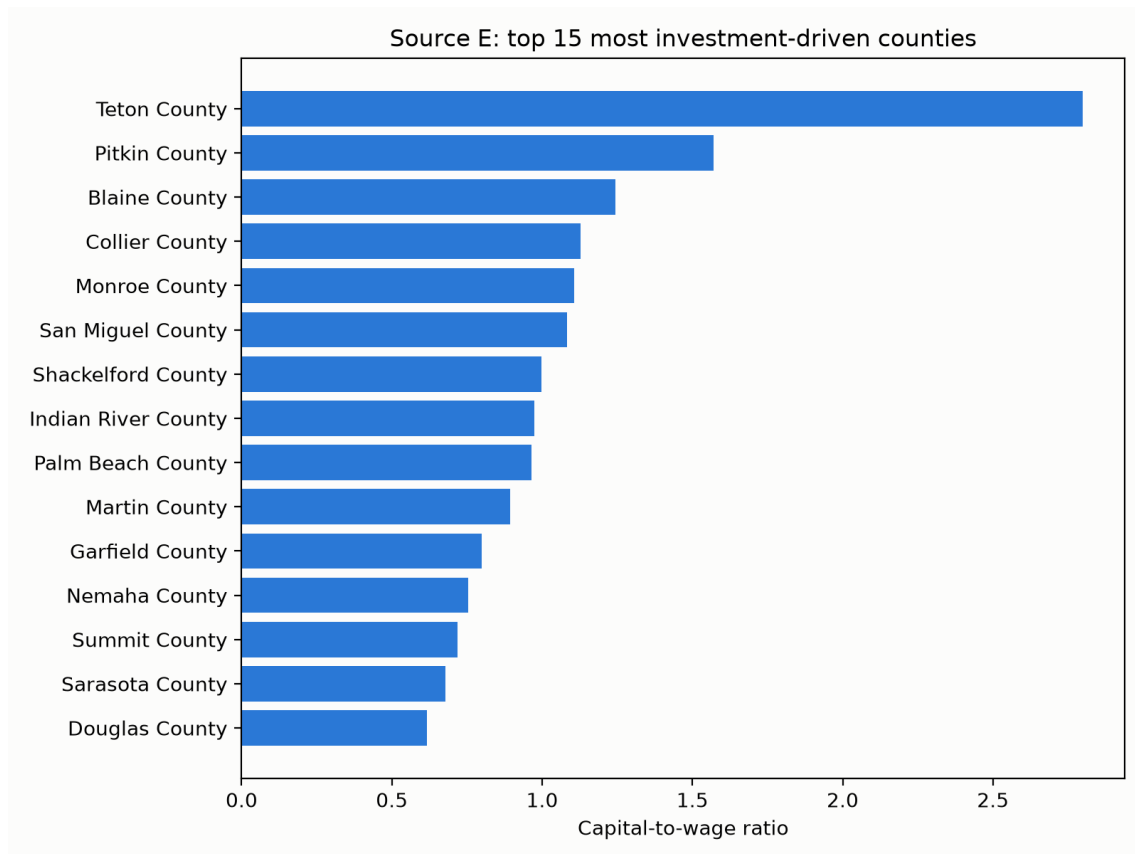


Top 15 counties by total freight tonnage. Harris, Los Angeles, and Cook Counties dominate by a wide margin — the roughly two-orders-of-magnitude spread between top hubs and the median county.

3.5 Source E — IRS Statistics of Income Panel Data (Capital Composition)

Source E computes `capital_to_wage_ratio` (net capital gains + qualified dividends, divided by wages) from IRS county-level tax aggregates, Tax Year 2022. Coverage is 3,143/3,144 (again, only Kalawao County, HI is absent — the same gap Source B independently hit). The top of the ratio distribution is dominated by well-known resort/second-home markets (Teton County, WY/Jackson Hole tops the list at 2.80, followed by Aspen, Sun Valley, Naples, the Florida Keys) — exactly the "asset-rich, investment-driven markets" the proposal describes. But the top 25 don't split cleanly into just resort-vs-labor: Benton County, AR (Walmart/Tyson/J.B. Hunt headquarters) is a large, non-resort county whose high ratio likely reflects concentrated executive/founder equity compensation, a third mechanism the proposal's framing doesn't name. A second real limitation: seven of the top-25 counties have well under 2,200 tax returns, and their high ratios look like single large one-time transactions (plausibly land sales) rather than sustained investment income — downstream consumers should weight by `num_returns` rather than trust the raw ratio uniformly. Unlike Source B, the IRS file carries no suppression flag at all, so a handful of ultra-low-population counties' exact-zero values can't be distinguished from suppressed ones. Against Source C, the ratio shows a small but real positive link to GDP velocity (monotonic across quartiles, 1.6% → 2.3%) and no significant link to unemployment velocity — directionally consistent with the proposal's own "tied to Wall Street, not local jobs" framing, though the effect sizes are small.

Proposal alignment: supported — the mechanism works exactly as described and checks out against real geography, with two new, undisclosed-by-either-scoping-doc limitations (no suppression flag; low-return-count noise) documented for downstream use.

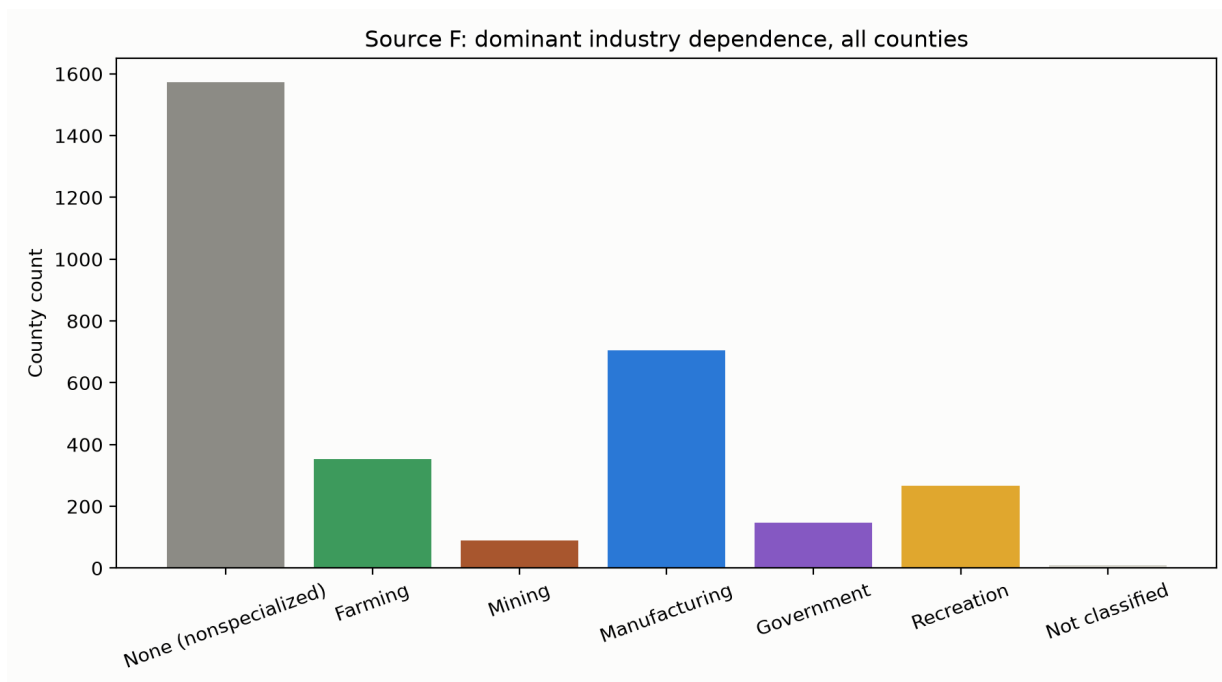


Top 15 counties by capital-to-wage ratio. Dominated by well-known resort/second-home markets (Teton/Jackson Hole, Pitkin/Aspen, Blaine/Sun Valley, Collier/Naples, Monroe/Florida Keys) — a direct visual check against real-world wealth geography.

3.6 Source F — USDA ERS County Typology Codes (Structural Resilience Baseline)

Source F pulls USDA's 2025-edition county typology codes — a single static file requiring no API key or rate limiting, unlike every other source. Coverage is complete (3,144/3,144). The proposal frames this source as sorting counties into five dependency categories (Farming, Mining, Manufacturing, Government, Recreation), but half of all counties (50.0%) are "Nonspecialized" — more than double the next-largest category (Manufacturing, 22.5%) — a residual the proposal's framing undersells. All 9 counties carrying ERS's "not classified" sentinel are Connecticut's Planning Regions, the exact same root cause (Connecticut's 2022 county abolition) that independently produces Source C's missing-GDP gap — two unrelated federal agencies hitting the same data-lag artifact. Nonmetro counties carry meaningfully more compounding demographic distress than metro counties (mean distress count 1.069 vs. 0.659, a ~62% relative gap). Cross-validated against Source C, demographic distress shows essentially zero correlation with either velocity axis ($r = 0.007$, $r = -0.019$) — a genuinely informative negative result: Source F captures slow-moving structural character, not short-run momentum, and the two pillars should be read as complementary rather than redundant.

Proposal alignment: supported operationally (stable, low-maintenance baseline anchor, exactly as described) but complicated substantively — "Nonspecialized" is the modal category, not a minor residual, and the proposal's stated Trade Logistics and Capital Flow synergies are addressed directly in §4.



Seven-way industry-dependence breakdown. "None (nonspecialized)" is more than double the next-largest category (Manufacturing) — the proposal's five-dependency framing undersells how large this residual actually is.

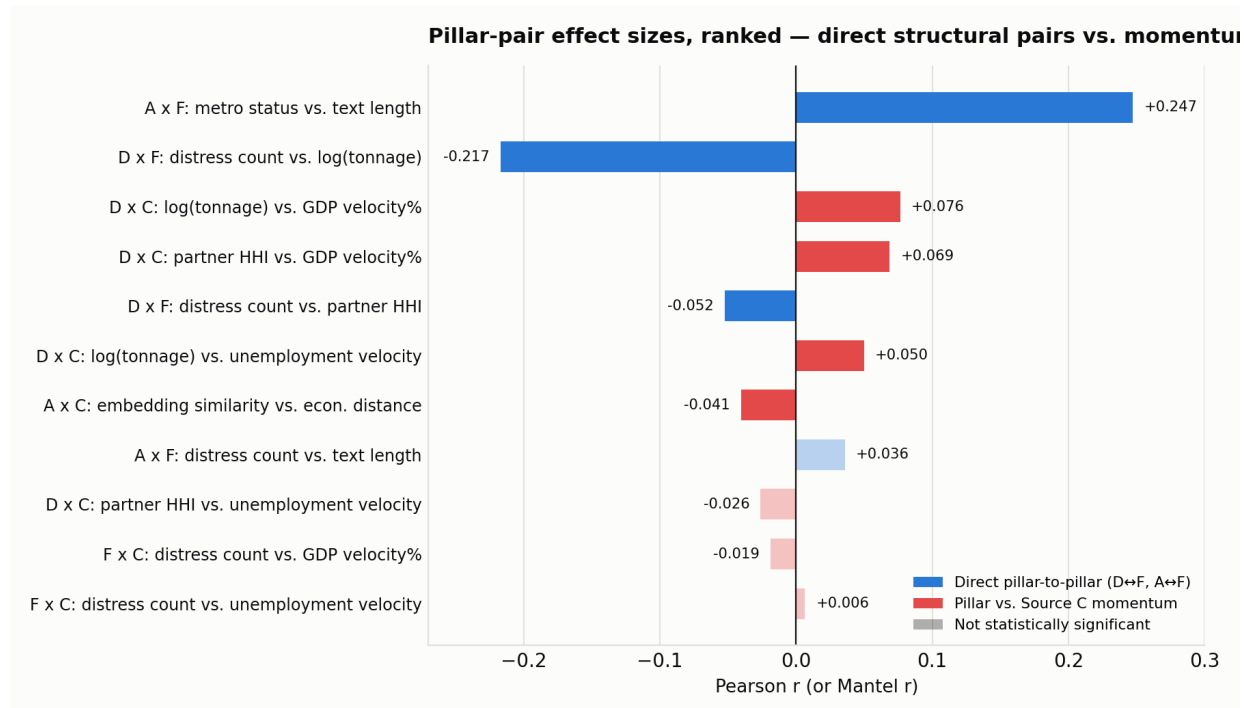
4. Cross-Source Synthesis

Two rounds of cross-source analysis were run to test the proposal's stated pillar synergies and close gaps each individual source's report left open.

Source A × Source C (closing Source A's own open item): testing embedding similarity against a *real economic variable* for the first time — rather than only geography — finds a Mantel r of -0.041 ($p = 0.002$, $n = 2,786$). Statistically real, but roughly 6x weaker than Source A's already-"weak" geography association, and too small to support the proposal's claim that intro text carries usable economic-transition signal. The ranked "textually similar but economically diverging" pair list is dominated by the same handful of corpus-extreme counties (Massac County, IL; Kittson County, MN; Campbell County, SD; Borden County, TX) — a known ranking artifact, not a set of genuinely informative pairs, and it mirrors an identical outlier-domination pattern Source C's own report found with King County, WA.

Three-pillar momentum crossvalidation (Sources A, D, F each against Source C, plus two new direct pillar-to-pillar tests): every pillar-vs-Source-C-momentum link is weak, now confirmed by permutation tests rather than bare point estimates ($|r| \leq 0.08$ in all cases; Source D's tonnage/HHI signals are the most consistently significant of the three, Source F's demographic-distress correlations are confirmed as statistically indistinguishable from noise). The two new direct pillar-to-pillar tests are the strongest results in the whole project: Source F's demographic distress count correlates with Source D's log-tonnage at $r =$

-0.217 ($p = 0.002$) — the proposal's Trade Logistics synergy, confirmed at moderate strength with a sensible ordering by industry-dependence category (Farming lowest, Manufacturing/Nonspecialized highest) — and Source F's metro status correlates with Source A's intro-text length at $r = 0.247$ ($p = 0.002$), the single largest effect size found anywhere in this project — the proposal's Capital Flow synergy, confirmed at the coarse metro/nonmetro level. A finer-grained version of Capital Flow does not hold: Source F's graded 0–6 distress count does not track Source A's text length beyond what the metro/nonmetro split already explains ($r = 0.036$, $p = 0.052$, wrong-signed relative to the proposal's implied direction), and this may simply reflect residual metro/nonmetro composition rather than an independent distress effect.



All 11 pillar-pair correlation tests run in this project, ranked by effect size. Direct pillar-to-pillar tests (blue) dwarf every test against Source C's momentum measure (red); faded bars are not statistically significant.

The chart above ranks all 11 pillar-pair tests run in this project by effect size. The two direct pillar-to-pillar tests (blue) are the clear outliers at the top; every test against Source C's momentum measure (red) — regardless of which structural pillar is doing the explaining — sits well below them, several failing to clear statistical significance at all. The pattern is consistent, not a fluke of one noisy test: momentum is a fundamentally harder target for any of these structural pillars to explain than another pillar's own static character is.

A separate, unrelated finding recurs across both rounds: Source C's 9 GDP-missing Connecticut geographies and Source F's 9 "not classified" sentinel counties are the exact same 9 Planning Regions — independent confirmation of a genuine upstream federal-data-lag issue, not a modeling result.

5. Recurring Data-Quality Findings

Several data-quality patterns surfaced independently across multiple sources, rather than being specific to any one pipeline:

- **Kalawao County, HI** (population ~90, the former Kalaupapa settlement) is absent from Source A (originally, later backfilled), Source B, Source C, and Source E's underlying provider data — a genuine, cross-source-consistent absence for the smallest county in the US, not a pipeline defect in any one source.
- **Connecticut's 2022 Planning-Region transition** independently breaks Source C's GDP series (all 9 regions missing) and Source F's industry-dependence classification (all 9 regions carry a "not classified" sentinel) — two unrelated federal agencies (BEA/FRED and USDA/ERS) with the same backfill lag.
- **Virginia's independent-city structure** required a purpose-built lookup table for Source A's Wikipedia article titles (56 entries total across two backfill rounds) and independently accounts for 51 of Source C's 64 GDP-coverage gaps — the same edge case surfacing in two unrelated ingestion pipelines.
- **Suppression handling differs meaningfully between sources that share the same underlying privacy problem:** Source B's BLS data ships an explicit disclosure flag (allowing honest nulls), while Source E's IRS data has no equivalent flag at all, leaving a small number of ultra-low-population counties' true zeros indistinguishable from suppressed values.

6. Consolidated Proposal Alignment Scorecard

Source	Core mechanism	Verdict
A	Intro-text embeddings carry distinctive economic-transition narrative	Not supported — strongest structure is a non-economic Wikipedia editorial artifact; economic correlation confirmed negligible in §4
B	Location Quotients identify which industries drive a county's economy	Supported — mechanism works as described; sector identity (not magnitude) predicts growth direction
C	Velocity quadrants differentiate fast-moving from slow-moving local economies	Partially supported — quadrant mechanism works, but a shared macro trend dominates over idiosyncratic local momentum; GDP velocity metric confounded with economy size
D	Freight flows distinguish pass-through corridors/exporters from consumer sinks	Supported at national scale, with two documented scope deviations (no matrix-explosion problem to solve; no dollar-value column)
E	Capital-to-wage ratio isolates asset-rich, investment-driven counties	Supported — mechanism checks out against real geography; two new limitations (no suppression flag, low-N noise) disclosed
F	Typology sorts counties into five mutually exclusive economic dependencies	Supported operationally, complicated substantively — "Nonspecialized" is the modal category, not a minor residual

Source	Core mechanism	Verdict
A × F (Capital Flow synergy)	Source F anchors counties where Source A's text is sparse	Confirmed at coarse (metro/nonmetro) level , not at fine-grained distress-count level
D × F (Trade Logistics synergy)	Source F's typology explains Source D's freight character	Confirmed at moderate strength , with a sensible industry-dependence ordering
All pillars vs. Source C	Each pillar helps explain Source C's short-run momentum	Not supported — every tested link is weak (

7. Global Limitations & Open Items

- **Source A ↔ Source D remains the one pillar pair never directly tested** — no specific proposal claim motivates it, so it wasn't prioritized, but it's the only pair missing from the cross-source matrix.
- **The A × F distress-count correlation may be confounded with metro status** — nonmetro counties independently have higher distress, so the borderline $r=0.036$ result could be residual composition rather than an independent effect; a partial correlation controlling for metro status was not run.
- **Source C's GDP velocity is dollar-denominated, not percentage-based**, in its originally-shipped column — every downstream cross-source test worked around this with a locally-computed `gdp_velocity_pct`, but the base parquet file itself has not been amended.
- **Source E has no suppression flag**, unlike Source B's equivalent BLS data — a structural gap in the source, not a pipeline bug, affecting a small, identified set of ultra-low-population counties.
- **Sample sizes differ across pillar pairs** — Source A applies its own stub-county and non-50-state filters ($n \approx 2,786\text{--}2,849$), while the other sources are at or near full 3,144-county coverage — so cross-pair effect-size comparisons in §4 are only approximately apples-to-apples.
- **Every cross-source correlation test is a single-round snapshot** against current data vintages (2022-2025 depending on source) — none of this round's findings have been checked for stability across time.

8. Consolidated Next Actions

- 1 Run the one remaining untested pillar pair, Source A × Source D, for completeness.
- 2 Run a partial correlation (or nonmetro-only subset) of Source F's distress count against Source A's text length, controlling for metro status, to resolve whether the Capital Flow synergy's fine-grained version is real or confounded.
- 3 Add a percentage-based GDP velocity column to Source C's base parquet file directly (rather than only computing it ad hoc in downstream scripts), since every cross-source test already needed it.
- 4 Cross-validate Source B's dominant sector directly against Source F's industry-dependence category (an agreement check between two independently-sourced industrial-structure signals) — flagged by both sources' own next-actions but not yet run.

- 5 Consider a `num_returns`-weighted variant of Source E's capital-to-wage ratio to directly address the low-return-count noise documented in §3.5, rather than leaving it as a caveat for downstream consumers.
- 6 Rebuild the "textually similar but economically diverging" pair lists (Source A × Source C, Source C's own King County analysis) on a percentile-rank or log-scaled distance metric to avoid the repeated single-outlier domination pattern found independently in both.